

Please note that there are tables of ΔH_f and bond energy values at the end of the exam. Please use these as needed.

1. A 50.0 g sample of copper at 125 °C is submerged in 200. g of water at 25.0 °C. What is the final temperature of the copper given the specific heat capacities below?

Specific heat capacities: Cu = 0.385 J/g • °C H₂O = 4.18 J/g • °C

- A) 22.6 °C
B) 27.3 °C
C) 75.0 °C
D) 127 °C
E) There is not enough information to solve this problem.
2. A given process needs 1.10×10^5 kJ of energy. All of the energy comes from the reaction of gaseous methane, CH₄, with O₂(g) to form gaseous water and gaseous CO₂. How many kg of methane are required to provide the energy needed?
- A) 1.77 kg
B) 1.94 kg
C) 2.20 kg
D) 2.49 kg
E) 5.50 kg
3. During a certain process, the volume of a cylinder with a piston decreases from 2.50 L to 1.20 L against a constant external pressure of 0.950 atm while 225 J of energy are lost as heat. Which statement about this process is true?
- A) $\Delta E = \Delta H$ for this process.
B) ΔH for this process is 0 because it occurs at constant pressure.
C) ΔE for this process cannot be determined because it does not occur at constant volume.
D) During this process, the system loses 1.24 J of energy as work.
E) ΔH for this process is -225J.

4. The combustion of methane, CH_4 , is represented by the chemical reaction $\text{CH}_4(\text{g}) + 2 \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2 \text{H}_2\text{O}(\text{g})$. Use bond energies to determine the enthalpy of reaction for the combustion.
- A) -892 kJ
B) -802 kJ
C) -586 kJ
D) 194 kJ
E) 854 kJ
5. Ultraviolet (UV) radiation often is classified as UV-A (longer wavelength) and UV-B (shorter wavelength). Both types of UV radiation contribute to skin cancer, and UV-B radiation is primarily responsible for severe sunburns. Calculate the energy (in kJ) of *one mole* of UV-B photons with wavelength 295 nm.
- A) 6.74×10^{-16} kJ
B) 1.18×10^{-13} kJ
C) 0.406 kJ
D) 0.674 kJ
E) 406 kJ
6. The Paschen series refers to the series of hydrogen emission spectra lines that appear in the infrared region. Each transition ends in the $n = 3$ energy level. The first three lines in this series are recorded with wavelengths of 1094 nm, 1282 nm, and 1875 nm. Assign the initial value of n for each wavelength.

1094 nm	1282 nm	1875 nm
A) $n = 1$	$n = 2$	$n = 3$
B) $n = 4$	$n = 5$	$n = 6$
C) $n = 5$	$n = 3$	$n = 2$
D) $n = 6$	$n = 5$	$n = 4$
E) $n = 3$	$n = 4$	$n = 5$

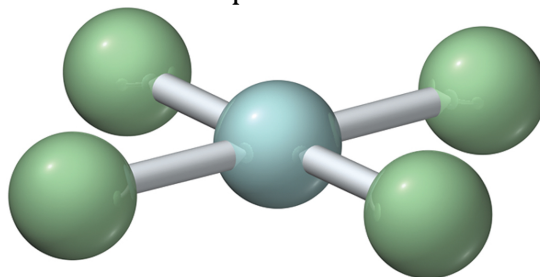
7. How does the p_z orbital in the $n = 4$ shell compare with the p_z orbital in the $n = 2$ shell in the same atom?
- A) The m_l values are the same.
 - B) The orientation of the orbital in the $n = 4$ shell is rotated 90° relative to the orbital in the $n = 2$ shell.
 - C) The two orbitals have different shapes.
 - D) The value of l is larger for the orbital in the $n = 4$ shell.
 - E) The two orbitals have the same number of radial nodes.
8. Which orbital does the last electron of cesium (Cs) occupy in its ground state?
- A) $4d$
 - B) $4f$
 - C) $5d$
 - D) $6s$
 - E) $6p$
9. How many unpaired electrons are there in the ground state electron configuration for Fe^{3+} ?
- A) 0
 - B) 1
 - C) 2
 - D) 3
 - E) 5
10. Which statement is true regarding the effective nuclear charge experienced by the valence electrons of beryllium (Be) relative to that experienced by the valence electrons of nitrogen (N).
- A) The valence electrons of N experience smaller effective nuclear charge because there are more valence electrons and the same number of core electrons as for Be.
 - B) The valence electrons of N experience greater effective nuclear charge because there are more protons but the same number of core electrons as for Be.
 - C) The valence electrons of Be experience greater effective nuclear charge because there are fewer protons and valence electrons and the same number of core electrons as for Be.
 - D) The valence electrons of Be experience smaller effective nuclear charge because it is a smaller atom.
 - E) The valence electrons of both atoms experience the same effective nuclear charge because they have the same number of core electrons.

11. Consider the set of successive ionization energies given below. How many valence electrons are there for the element to which the ionization energies belong?

$$\begin{aligned} \text{IE}_1 &= 578 \text{ kJ/mol} \\ \text{IE}_2 &= 1820 \text{ kJ/mol} \\ \text{IE}_3 &= 2750 \text{ kJ/mol} \\ \text{IE}_4 &= 11,600 \text{ kJ/mol} \end{aligned}$$

- A) 1
B) 2
C) 3
D) 4
E) 5
12. Arrange the ionic compounds in order of *increasing magnitude* (absolute value) of lattice energy: CaO, KBr, KCl, SrO.
- A) $\text{KBr} < \text{KCl} < \text{SrO} < \text{CaO}$
B) $\text{CaO} < \text{SrO} < \text{KCl} < \text{KBr}$
C) $\text{KCl} < \text{KBr} < \text{CaO} < \text{SrO}$
D) $\text{SrO} < \text{CaO} < \text{KBr} < \text{KCl}$
E) $\text{KCl} = \text{KBr} < \text{CaO} = \text{SrO}$
13. Draw the Lewis structure for SCl_2 and select the answer that best describes it.
- A) There are two single bonds between Cl and S, and there are two lone pairs on the central atom.
B) There are two double bonds between Cl and S, and there is one lone pair on the central atom.
C) There are two double bonds between Cl and S, and there are two lone pairs on the central atom.
D) There are two single bonds between Cl and S, and there are three lone pairs on the central atom.
E) There are two triple bonds between Cl and S, and there are no lone pairs on the central atom.

14. Which bond is most likely to be classified as a polar covalent bond?
- A) C—C
 - B) a bond between a metal and nonmetal
 - C) the bonds in a sample of Na(s)
 - D) C—H
 - E) C—O
15. In the laboratory, a student recreates the experiments of the photoelectric effect. Which observation correctly correlates to the experiments?
- A) Electrons are ejected from the metal surface at all frequencies of the incident light.
 - B) The number of electrons ejected increases linearly with the frequency of the incident light, as long as the frequency of the light is above a certain value.
 - C) The kinetic energy of the ejected electrons increases linearly with the intensity of the incident light as long as the frequency of the light is above a certain value
 - D) The number of electrons ejected increases linearly with the intensity of incident light, as long as the frequency of the light is above a certain value.
 - E) The kinetic energy of ejected electrons is independent of the frequency of incident light.
16. Determine the number of total electron groups, the number of bonding groups, and the number of lone pairs on the central atom for the structure:



	Total electron groups	Bonding groups	Lone pairs
A)	4	4	0
B)	5	4	1
C)	4	0	4
D)	5	4	0
E)	6	4	2

17. For which molecule(s) do you expect deviations from ideal bond angles?

- I. CF_4
- II. OF_2
- III. ClF_3

- A) I only
- B) II only
- C) III only
- D) Both I and III
- E) Both II and III

18. Which molecule contains polar bonds but has no net dipole moment?

- I. CCl_4
- II. PCl_3
- III. ICl_4^-

- A) I only
- B) II only
- C) III only
- D) Both I and II
- E) Both I and III

19. Which answer option best describes the sigma bonds in NH_3 ?

- A) The sigma bond is formed by overlap in the internuclear axis of an sp^2 hybrid orbital on N and a 1s atomic orbital on H.
- B) The sigma bond is formed by overlap both above and below the internuclear axis of a 2p atomic orbital on N and a 1s atomic orbital on H.
- C) There are no sigma bonds in NH_3 .
- D) The sigma bond is formed by overlap in the internuclear axis of an sp^3 hybrid orbital on N and a 1s atomic orbital on H.
- E) The sigma bond is formed by overlap above and below the internuclear axis of an sp^2 hybrid orbital on N and a 1s atomic orbital on H.

20. Given the diagram below, which statement is *true*?

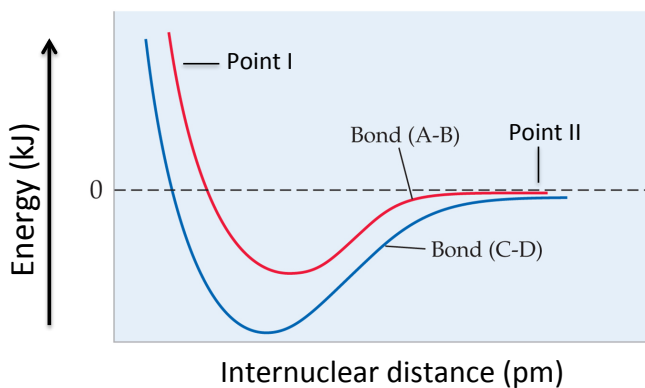


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- A) Point I on the graph indicates the energy released when bond A-B (or bond C-D) is formed.
- B) Point II on the graph indicates the internuclear distance at which atoms A and B (or atoms C and D) repel one another.
- C) **Bond C-D is shorter and stronger than bond A-B**
- D) More energy is released when bond A-B is formed than when bond C-D is formed.
- E) Point II indicates that the two bonds have the same bond dissociation energy.

21. Which form of the exam do you have?

- A)
- B)

Substance	ΔH_f (kJ/mol)
C(s, graphite)	0
CH ₄ (g)	-74.6
CO ₂ (g)	-393.5
CO ₂ (aq)	-413.8
H(g)	218.0
H ₂ (g)	0
O ₂ (g)	0
H ₂ O(l)	-285.8
H ₂ O(g)	-241.8

Bond	Average Bond Energy (kJ/mol)
H—H	436
H—C	414
H—O	464
C—C	347
C=C	611
C—O	360
C=O in CO ₂	799
O—O	142
O=O	498